PRATHAM

"Pre-hospital intelligence. Post-arrival readiness."

A presentation by Code Pirates

Team members:

- Rama Sanjana
- Vinay Boppidi
- Kalva Maanya

CODE PIRATES

HACKING THE DIGITAL SEAS

PROBLEM STATEMENT

AI Triage Optimizer for Government Hospitals

Every minute of delay in a government hospital emergency room can cost a life. Yet today, most OPDs in India still rely on manual, first-come-first-served queues — with no system to identify who needs urgent care first.

The Scale of the Problem	What's Broken Today	🔴The Human Cost
• India has 25,000+ government hospitals serving 150M+ OPD patients annually • Doctor-to-patient ratio in government hospitals: 1:2000 (WHO recommends 1:1000) • Average OPD wait time: 2–4 hours, regardless of severity	• No triage system at most government OPD entry points • Nurses make prioritization calls manually, under extreme pressure • Critical symptoms go undetected until the patient deteriorates • No audit trail — decisions are undocumented	• Cardiac events, strokes, and sepsis are time-critical — every 10 minutes matters • Studies show structured triage reduces critical care mortality by up to 30% • Most deaths in OPD waiting areas are preventable with early identification

SOLUTION: PRATHAM

No guesswork. No bias. No delay.

Captures 6 Vitals
Critical vitals + symptoms at OPD entry in under 2 minutes.

Assigns Priority
Instantly with structured reasoning and confidence score.

Predicts Deterioration
Predicts high-risk cases for the next 30 minutes.

Real-time Alerts
'Cardiac event risk detected, immediate attention required'

Ambulance Mode
Minimal vitals Entered on move, hospital prep alert.

Nurse Override
AI assists but human always has final authority.

Live Dashboard
Real-time counts, wait times, and triage summary.

Pratham — System Architecture

AI Triage System for Emergency Departments

USERS

- Nurse / Doctor
- Paramedic
- Administrator

LEGEND

- Frontend (React)
- Backend (Node/Express)
- Database (Supabase)
- Data Flow

FRONTEND — React 18 + TypeScript + Vite (localhost:8080)

PAGES
- TriageInput.tsx
- TriageResult.tsx
- Dashboard.tsx
- NotFound.tsx

COMPONENTS
- ExplainabilityPanel
- RiskAssessmentCard
- AmbulanceModeToggle
- AppHeader · VitalInput

i18n
- English
- Hindi
- Telugu

STATE & ROUTING
- React Router v6
- TanStack Query
- Axios + Interceptors
- Priority Mapper

UI LAYER
- Tailwind CSS
- shadcn/ui Components
- Recharts (Dashboard)
- Lucide React Icons

HTTP / REST

BACKEND — Node.js + Express + TypeScript (localhost:3000)

API ROUTES
- POST /api/triage
- GET /api/triage/:id
- POST /api/triage/:id/override
- GET /api/dashboard/summary
- GET /api/export/csv
- GET /api/audit-logs

AI TRIAGE ENGINE
- Vital Signs Scoring
- Symptom Scoring
- RED Flag Detection
- Priority: RED/YEL/GRN
- Confidence: 70–98%
- Score 0–100 pts

EXPLAINABILITY
- Reasoning Generator
- Why not CRITICAL?
- Why not STABLE?
- Bias Transparency
- Ethics Statement

RISK ASSESSMENT
- Deterioration: H/M/L
- Timeframe: 5min–2hr
- Risk Indicators
- Prep Checklists
- Team Alerts

AUDIT & EXPORT
- Winston Logger
- Audit Trail (Supabase)
- CSV Exporter
- Zod Validation
- Error Middleware

Supabase JS

DATABASE — Supabase · PostgreSQL · Row Level Security

triage_records patient · vitals · symptoms · priority · score · explain

audit_logs action · triage_id · details · operator

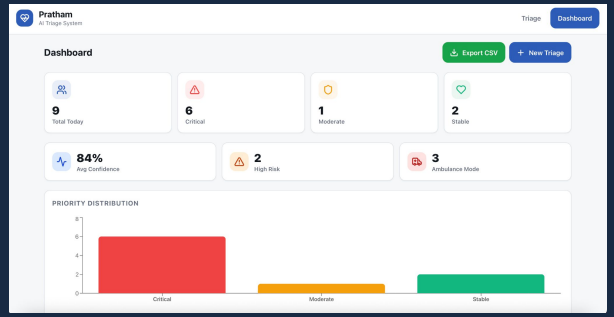
GET /health status · timestamp · version

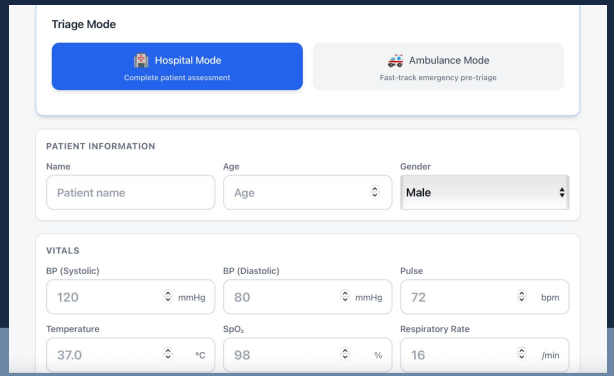
HOW IS OUR SOLUTION BETTER THAN THE EXISTING SOLUTIONS?

Aspect	🔴Existing Systems (Static Classification)	🟢Proposed System (Predictive Intelligence)
Core Approach	❌Classifies patient's *current condition* only	✅Predicts *future deterioration risk* (next 30 mins)
Prediction Capability	❌No prediction, only reactive	✅Proactive alerts (cardiac, respiratory risks)
Timing of Action	❌Action taken *after symptoms worsen*	✅Early intervention *before critical deterioration*
Nurse Accountability	❌No audit trail	✅Every decision timestamped
Ambulance Integration	❌Not integrated with pre-hospital care	✅Starts triage *inside ambulance* (Ambulance Mode)

HOW IS OUR SOLUTION BETTER THAN THE EXISTING SOLUTIONS?

Aspect	🔴Existing Systems (Static Classification)	🟢Proposed System (Predictive Intelligence)
Hospital Preparation	❌No prior alert to hospital	✅Sends *ETA-based alerts* to prepare ICU/Oxygen/Trauma
Handling High Patient Load	❌Struggles during OPD rush	✅Designed for *high patient volume* scenarios
Patient Prioritization	❌Based on visible symptoms only	✅Intelligent *risk-based prioritization*





TECH-STACK

BACKEND

- **Framework:** Node.js + Express + TypeScript
- **Database:** Supabase (PostgreSQL)
- **AI Engine:** Claude API (Anthropic) — triage reasoning + explainability
- **Authentication:** Supabase Auth — role-based nurse / doctor login
- **API Style:** REST API with structured JSON responses
- **Environment:** dotenv for secure credential management
- **Deployment:** Railway / Render cloud hosting

FRONTEND

- **Framework:** React + TypeScript + Vite
- **UI:** Tailwind CSS + shadcn/ui components
- **State:** React Hooks
- **Routing:** React Router v6
- **API:** Axios with priority mapping
- **i18n:** react-i18next with 3 locales

AI & DATA LAYER

- **AI Model:** Claude API (claude-sonnet-4-20250514)
- **Triage Logic:** Vitals + symptoms → RED / YELLOW / GREEN
- **Explainability:** Structured reasoning with confidence score
- **Risk Engine:** 30-minute deterioration prediction
- **Bias Control:** Zero demographic data in AI decision pipeline
- **Ambulance Mode:** Lightweight input → hospital prep alert

DEVOPS & TOOLS

- **Version Control:** GitHub — collaborative multi-device workflow
- **Frontend Builder:** Lovable — AI-generated UI
- **DB Management:** Supabase Dashboard + SQL Editor
- **Testing:** Postman for API validation
- **Project Tracking:** Notion / GitHub Issues

Integration Roadmap

Phase 1 — Pilot (Month 1–3)

- One district government hospital
- Single tablet at OPD entry
- Run alongside existing manual process
- Measure AI accuracy vs nurse decisions

Phase 2 — Validation (Month 3–6)

- Expand to 2–3 counters in same hospital
- Onboard nurses with 30-minute training
- Get CMO sign-off for formal adoption

Phase 3 — Scale (Month 6+)

- Apply to NHM for state-wide rollout
- Integrate with ABDM / ABHA ID
- Expand to multiple states

CONCLUSION

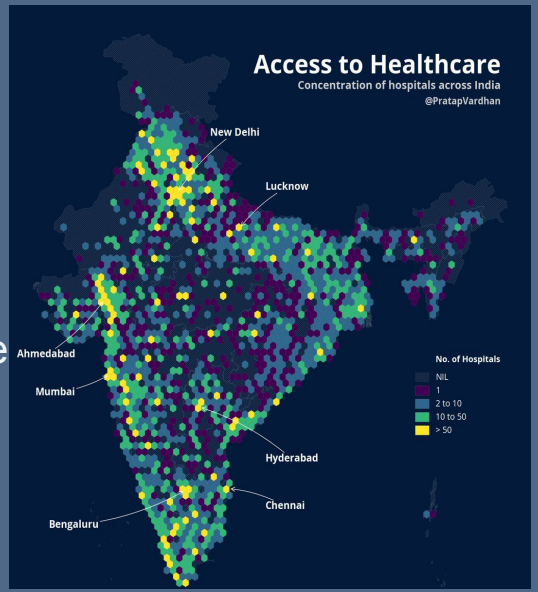
Pratham
AI Triage System

What It Changes

- A critical patient is identified in 2 minutes, not 45
- A nurse makes a confident, data-backed decision — not a pressured guess
- A hospital receives an ambulance alert before the patient arrives

The Bigger Picture

- 150M+ OPD patients annually — no standardized triage system exists today
- Sanjeevani is plug-and-play — one tablet, zero infrastructure change 30-minute onboarding
- Scalable from one OPD counter to a national rollout via NHM and ABDM



"One tablet. Two minutes. One life saved."